Grade 6 Lesson Plan: Arduino SPI & SD Cards

Lesson Overview

Target Audience: Grade 6 Students

Topic: How Arduino communicates with devices using SPI, specifically focusing on the SD Card Module.

Objective: Students will understand the basics of Arduino communication, identify the 4 pins of SPI, correctly wire an SD card module, and comprehend the basic code structure for writing to an SD card.

1. Introduction to Arduino & Communication

- **What is Arduino?** A tiny computer/brain that controls electronics.
- **Why Talk?** Devices need to share data to function together (e.g., reading a temperature sensor and saving it). Compare this to friends sharing secrets with walkie-talkies.

2. The Secret Language: SPI

SPI stands for Serial Peripheral Interface. It is a super-fast way for Arduino to talk clearly with devices like SD cards.

The 4 Magic Wires of SPI:

- **MOSI (Master Out, Slave In):** Arduino TALKS and sends data to the device.
- **MISO (Master In, Slave Out):** Arduino LISTENS and receives data from the device.
- **SCK (Clock):** Keeps perfect TIME like a drumbeat so messages don't get mixed up.
- **SS (Slave Select):** CHOOSES which device to talk to. Important when multiple devices share the same MOSI, MISO, and SCK lines.

3. The SD Card Module

SD cards act as a digital notebook for the Arduino, allowing us to store thousands of measurements, songs, or game scores.

Wiring Guide:

- **MOSI:** Pin 11
- **MISO:** Pin 12
- **SCK:** Pin 13
- **CS** / **SS:** Pin 10
- **VCC:** 5V
- **GND:** GND

4. Let's See Some Code!

To talk to the SD card, we need special libraries to help Arduino understand.

```
#include <SPI.h>
#include <SD.h>

void setup() {
  SD.begin(10); // Start SD card on Pin 10
  File myFile = SD.open("hello.txt", FILE_WRITE);
  myFile.println("Hello World!");
  myFile.close();
}
```

- `SPI.h`: Lets Arduino talk fast.
- `SD.h`: Helps read and write files.
- The process: Start SPI -> Open File -> Write/Read Data -> Close.

5. Real World Projects

- **Data Logger:** Record weather data (temp/humidity) over time.
- **Music Player:** Play songs stored on the SD card.
- **Game Score Keeper:** Save high scores.
- **Camera Module:** Capture and save photos.